Information Visualization: Assignment 2

Veneta Grigorova

November 2026

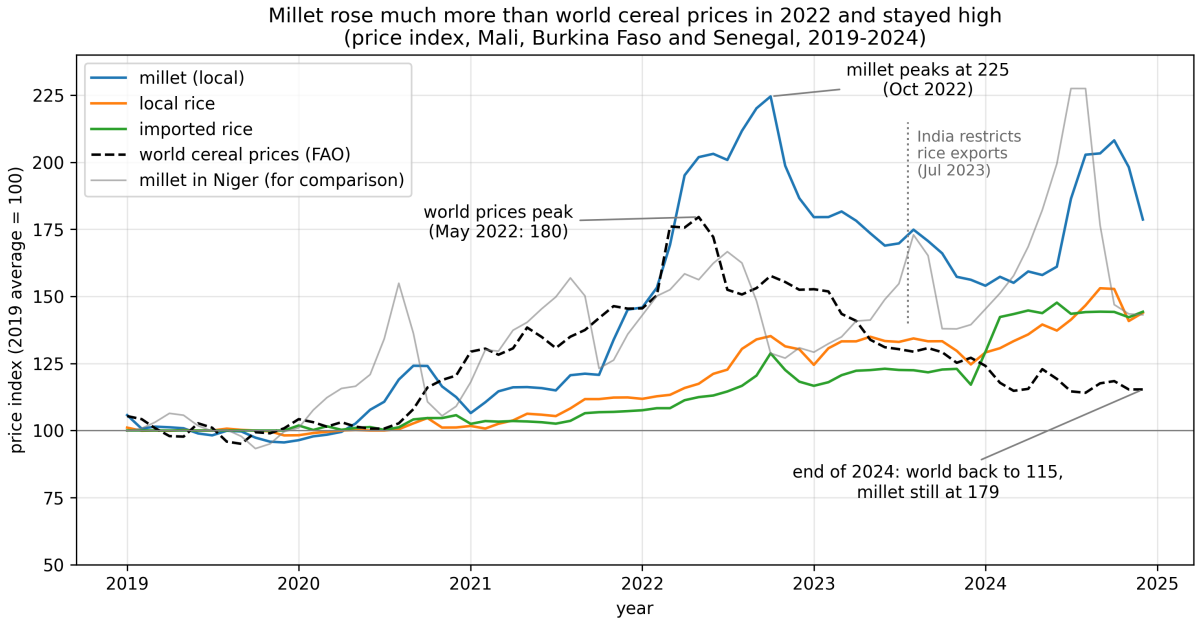
1 Research Question

Globally, food prices increased a lot in 2022. I wanted to see if local markets in the Sahel followed this trend. I asked two questions: "Did local food prices go up and down together with world cereal prices between 2019 and 2024?" and "How much did each food rise from 2021 to 2022 compared to world cereal prices, and is it the same in each country?". My hypothesis was that if world prices caused the rise, imported rice should increase the most. If not, local prices would follow their own pattern.

I used our team's cleaned WFP price data [1] and the FAO Food Price Index [2]. The WFP data is published on the Humanitarian Data Exchange. WFP staff go to local markets every month and record the prices of staple foods. Each row is one price for one food in one market and month. I compared three foods: millet, which is grown locally, local rice and imported rice. I only used Mali, Burkina Faso and Senegal. Niger has no local rice at all, and Chad has it in only one market. A single market's price can't stand for a whole country, so in those two countries the three-way comparison isn't possible.

2 Visualization 1: Local and World Prices over Time

Question: *Did local food prices go up and down together with world cereal prices between 2019 and 2024?*



Source: WFP food prices (HDX), FAO Food Price Index. Median of all markets in the 3 countries (grey line: markets in Niger), retail prices, not adjusted for inflation.

2.1 Design rationale

A line chart is good for monthly data, because you can see when prices go up and when they come down. I didn't compare the foods in CFA francs because they have very different prices per kg. Instead, the y-axis is a price index where the 2019 average is 100. The grey line marks this 2019 level, so anything above it is more expensive than in 2019. I gave the FAO index the same base, so all lines can be compared. The world prices are a black dashed line because they are a reference not a food. Each food has its own colour. The title tells the reader the main finding, and the labels point out the key moments. I also marked July 2023 with a dotted line, when India restricted rice exports and world rice prices went up [3].

2.2 What the chart shows

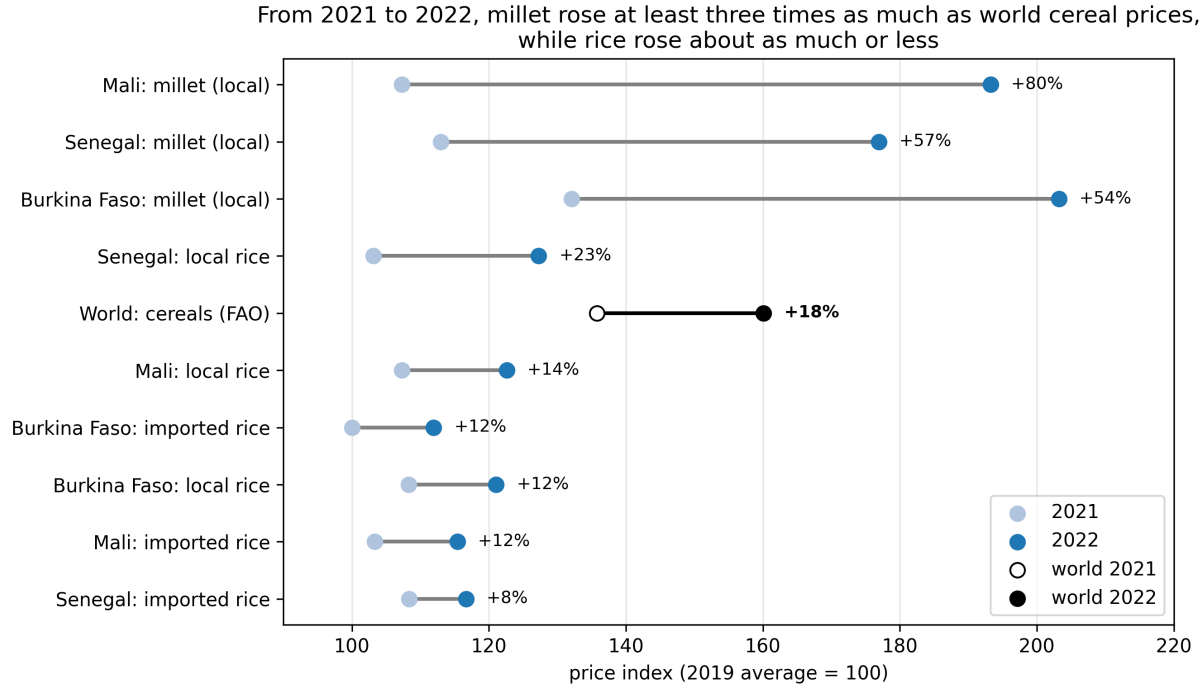
Local prices did not just follow world prices. In 2020 and 2021 world prices went up, but local prices barely moved. In early 2022 millet rose together with world prices, but then it went much higher. It peaked in October 2022, at the end of the lean season before the harvest. In 2024 millet rose again while world prices kept falling. Rice reacted slowly. Imported rice only jumped in early 2024, about half a year after India restricted rice exports. By the end of 2024 all local foods were still above the world index.

2.3 What the chart may obscure

Each line is the median of all markets in the three countries together. So the differences between countries are lost, and Mali counts more because it has more markets. The index hides the real prices, for example that rice costs more per kg than millet. The FAO cereals index is mostly wheat and maize, so it does not show world rice prices, which went up in 2023 when the cereals index fell. Finally, the chart shows that local and world prices moved differently, but it cannot show why. The reasons may be better explained by my teammates who look into factors like conflict and rainfall.

3 Visualization 2: Price Change from 2021 to 2022

Question: *How much did each food rise from 2021 to 2022 compared to world cereal prices, and is it the same in each of the three countries?*



Source: WFP food prices (HDX), FAO Food Price Index. Yearly median of all markets, retail prices.

3.1 Design rationale

The first chart puts the three countries together, so I wanted to look at each country and food separately. I chose 2021 and 2022 because 2022 is the year of the world price spike, and 2021 is the year just before it. The lighter dot is 2021 and the darker dot is 2022. A longer line means a bigger price increase. Each row is the yearly median of the price index (2019 = 100). I calculated the percent change for the reader and sorted the rows from the biggest to the smallest change. Because of this, the three millet rows end up at the top and the rice rows at the bottom, so the reader can see the result before reading any numbers. The world row in black works as a dividing line: rows above it rose more than world prices, rows below it rose less.

3.2 What the chart shows

The pattern is the same in all three countries. Millet rose by 54% to 80%, at least three times as much as world cereal prices (+18%). Imported rice rose the least, only 8% to 12%. This goes against my hypothesis: if world prices had caused the rise, imported rice should have risen the most. So millet was probably driven by regional conditions and not by world prices. Our team's data shows that conflict rose sharply in Mali and Burkina Faso in 2022, but Senegal had no conflict and millet still rose there. One possibility is that poor harvests in other parts of the Sahel, like Niger, raised millet prices across the region [4].

3.3 What the chart may obscure

The chart only compares two years, so it cannot show what happened before 2021 or after 2022. The yearly median hides changes within the year. The rows are based on very different numbers of markets, from 13 for imported rice in Burkina Faso to 61 for millet. And percent change depends on the starting level. Millet in Burkina Faso was already high in 2021, so its rise looks smaller than in Mali, even though its 2022 price was the highest.

References

[1] World Food Programme. Food prices for Mali, Burkina Faso and Senegal. Humanitarian Data Exchange, licence CC BY-IGO. https://data.humdata.org/dataset/global-wfp-food-prices. Accessed 24 October 2026.

[2] Food and Agriculture Organization of the United Nations. FAO Food Price Index. https://www.fao.org/worldfoodsituation/foodpricesindex/en/. Accessed October 2026.

[3] Global rice prices hit 15-year high after India's export ban: UN FAO. The Wire, 9 September 2023. https://thewire.in/world/global-rice-prices-hit-15-year-high-after-indias-export-ban-un-fao.

[4] FAO, FEWS NET and WFP. West and Central Africa: Market situation in 2021 and 2022 outlooks. February 2022. https://reliefweb.int/report/burkina-faso/west-and-central-africa-market-situation-2021-and-2022-outlooks-february-2022.